

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397784
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Onitsuka; Masakatsu

Vehicle control system

Abstract

A vehicle control system includes an on-vehicle sensor for outputting a piece of information regarding an object around an own vehicle and a piece of information regarding a travel state of the own vehicle, a driving assistant ECU which has, as a risk reduction function, at least one of a lane departure prevention function and an emergency stop function, and a switching apparatus for enabling and disabling the risk reduction function. The lane departure prevention function prevents departure of the own vehicle from a lane in which the own vehicle is travelling. The emergency stop function stops the own vehicle when a driver is suffering from an abnormal condition. In the case where the own vehicle collides with an object in a state in which the risk reduction function has been disabled by the switching apparatus, the driving assistant ECU forcibly enables the risk reduction function.

Inventors:	Onitsuka; Masakatsu (Nisshin, JP)
Applicant:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
Family ID:	1000008780574
Assignee:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
Appl. No.:	18/164869
Filed:	February 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230278549 A1	Sep. 07, 2023

Foreign Application Priority Data

JP	2022-032467	Mar. 03, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: B60W30/09 (20120101); B60R21/0136 (20060101); B60W30/12 (20200101); B60W40/08 (20120101)

U.S. Cl.:

CPC B60W30/09 (20130101); B60R21/0136 (20130101); B60W30/12 (20130101); B60W40/08 (20130101);

Field of Classification Search

CPC: B60W (30/09); B60W (30/12); B60W (40/08); B60W (2520/10); B60W (2520/105); B60W (2540/221); B60W (10/06); B60W (10/18); B60W (10/20); B60W (30/182); B60W (50/082); B60W (50/087); B60W (2050/143); B60W (50/14); B60W (10/10); B60W (30/085); B60W (2420/54); B60W (2510/10); B60W (2510/20); B60W (2540/10); B60W (2540/12); B60W (2710/06); B60W (2710/10); B60W (2710/18); B60W (2710/20); B60R (21/0136); B60T (7/12)

USPC: 701/36

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7692534	12/2009	Kataoka et al.	N/A	N/A
8352124	12/2012	Taguchi	N/A	N/A
8682500	12/2013	Sakugawa	N/A	N/A
8818634	12/2013	Fujita et al.	N/A	N/A
9168953	12/2014	Mitsumoto et al.	N/A	N/A
9714034	12/2016	Otake et al.	N/A	N/A
9880558	12/2017	Nakamura	N/A	N/A
9902399	12/2017	Torii et al.	N/A	N/A
10345443	12/2018	Masui et al.	N/A	N/A
10611240	12/2019	Masui et al.	N/A	N/A
2019/0077401	12/2018	Katagiri	N/A	B60W 10/18
2021/0016783	12/2020	Watanabe	N/A	B60W 30/18154
2022/0063667	12/2021	Wolff	N/A	B60W 60/0011
2022/0297541	12/2021	Aurand	N/A	B60W 50/10

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2012-252500	12/2011	JP	N/A

Primary Examiner: Hannan; B M M

Attorney, Agent or Firm: Sughrue Mion, PLLC

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a vehicle control system which has a function of enhancing safety of a vehicle.

Description of the Related Art

(2) There has been known a vehicle control system (hereafter referred to as the “conventional system”) which has a function of enhancing safety of an own vehicle (hereinafter referred to as a “risk reduction function”) (see, for example, Japanese Patent Application Laid-Open (kokai) No. 2012-252500). This conventional system has, as the risk reduction function, functions used when the own vehicle deviates from a widthwise center portion of a lane toward an edge (right edge or left edge) of the lane; i.e., a function of reproducing a predetermined warning sound, and a function of controlling a steering apparatus, a drive apparatus, a braking apparatus, etc. (through automatic steering, automatic braking, etc.) such that the own vehicle returns to the widthwise center portion of the lane (hereinafter, these functions are collectively referred to as a “lane departure prevention function”).

(3) Incidentally, in some cases, a vehicle control system having a risk reduction function as described above is configured to allow a driver to manually perform switching between a state in which the risk reduction function is enabled and a state in which the risk reduction function is disabled. In such a case, in a state in which the risk reduction function has been manually disabled, reproduction of a warning sound, automatic steering, automatic braking, or the like is not performed even when the own vehicle falls into a situation which is unfavorable in terms of safety.

SUMMARY OF THE INVENTION

(4) One object of the present invention is to provide a vehicle control system which can enhance safety of a vehicle.

(5) In order to solve the above-described problem, a vehicle control system (1) of the present invention comprises: an on-vehicle sensor (20) for outputting a piece of information regarding an object around an own vehicle and a piece of information regarding a travel state of the own vehicle; a control apparatus (10) which has, as a risk reduction function, at least one of a lane departure prevention function and an emergency stop function, the lane departure prevention function controlling at least one of a warning apparatus (24), a drive apparatus (30), a braking apparatus (40), and a steering apparatus (60) of the own vehicle, on the basis of the pieces of information obtained from the on-vehicle sensor, such that departure of the own vehicle from a lane in which the own vehicle is travelling is prevented, the emergency stop function stopping the own vehicle when a driver is suffering from an abnormal condition; and a switching apparatus (2b) for enabling and disabling the risk reduction function.

(6) The control apparatus is configured to forcibly enable the risk reduction function when the own vehicle collides with an object in a state in which the risk reduction function has been disabled by the switching apparatus.

(7) The switching apparatus is, for example, a switch which can be manually operated by a driver (occupant). Alternatively, the switching apparatus includes, for example, an apparatus which automatically disables the lane departure prevention function upon detection that the driver has intentionally performed a lane change and an apparatus which automatically enables the lane departure prevention function upon detection that the driver is suffering from an abnormal condition.

(8) In a vehicle on which the above-described vehicle control system is mounted, the risk reduction function is enabled and disabled by the switching apparatus. For example, when the own vehicle

departs from a lane and collides with an object in a state in which the risk reduction function has been disabled, the risk reduction function is forcibly enabled. As a result, for example, in the case where the control apparatus has the lane departure prevention function as the risk reduction function, in a situation where it has become difficult for the driver to drive and operate the own vehicle, the driving and operation are assisted by the control apparatus. Also, in the case where the control apparatus has the emergency stop function as the risk reduction function, in a situation where the driver has become unable to drive and operate the own vehicle, the control apparatus stops the own vehicle. Accordingly, the vehicle control system according to the present invention can enhance safety of the vehicle as compared with the conventional system.

(9) The control apparatus may determine that the own vehicle has lightly collided with an object in the case where the magnitude of acceleration (G) of the own vehicle falls within a predetermined range (GA) of acceleration greater than accelerations which may occur during an ordinary travel, and an integral value of the magnitude of acceleration of the own vehicle has exceeded a threshold value, the integral value of the magnitude of acceleration being that over a predetermined time from a point in time when the magnitude of the acceleration has exceeded a lower limit of the predetermined range.

(10) By virtue of this configuration, the risk reduction function can be forcibly enabled in a situation where the own vehicle has collided with the object relatively lightly (for example, a collision which does not cause an airbag apparatus to operate).

(11) The switching apparatus may include a first operating element (**2b1**) for manually enabling and disabling the lane departure prevention function and a second operating element (**2b2**) for manually enabling and disabling the emergency stop function, and, in the case where the own vehicle departs from the lane and collides with an object in a state in which at least one of the lane departure prevention function and the emergency stop function has been manually disabled, the control apparatus may forcibly enable the function disabled manually.

(12) This configuration allows the driver to enable part of the risk reduction functions, and disable the remaining function(s) according to the driver's preference.

(13) The risk reduction function may be the emergency stop function, and the control apparatus may control the braking apparatus so as to immediately stop the own vehicle, irrespective of whether or not the driver is suffering from an abnormal condition, in the case where the speed of the own vehicle exceeds a threshold value in a period between a point in time when the control apparatus has forcibly enabled the risk reduction function and a point in time when braking of the own vehicle by the emergency stop function is started.

(14) By virtue of this configuration, for example, in the case where the own vehicle hits against an object at a relatively high speed, the own vehicle can be stopped immediately.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of a vehicle control system according to one embodiment of the present invention;
- (2) FIG. 2 is a plan view showing a scene in which a vehicle has slightly deviated leftward from a widthwise center portion of a lane;
- (3) FIG. 3 is a time chart showing a process of execution of various types of controls when the own vehicle departs from a lane and lightly collides with an object;
- (4) FIG. 4 is a flowchart of a first program (lane departure prevention function);
- (5) FIG. 5 is a flowchart of a second program (emergency stop function); and
- (6) FIG. 6 is a flowchart of a third program (compulsory activation function).

DESCRIPTION OF THE PREFERRED EMBODIMENT

(7) (Outline of Configuration)

(8) As shown in FIG. 1, a vehicle control system **1** according to one embodiment of the present invention is mounted on a vehicle **V** (hereinafter referred to as the “own vehicle” in some cases). As will be described in detail later, the vehicle control system **1** has a risk reduction function which enhances safety of the own vehicle. The risk reduction function includes not only a lane departure prevention function which is the same as that of the above-described conventional system but also an emergency stop function of immediately stopping the own vehicle when a driver is suffering from an abnormal condition. The driver can manually perform switching between a state in which the risk reduction function is enabled and a state in which the risk reduction function is disabled; namely, the driver can manually enable and disable the risk reduction function. Furthermore, the vehicle control system **1** has a compulsory activation function of forcibly enabling the risk reduction function when a predetermined condition is satisfied.

(9) (Specific Configuration)

(10) As shown in FIG. 1, the vehicle control system **1** includes a driving assistant ECU **10**, on-vehicle sensors **20**, a drive apparatus **30**, a braking apparatus **40**, a shift change apparatus **50**, and a steering apparatus **60**.

(11) The driving assistant ECU **10** includes a microcomputer having a CPU **10a**, a ROM **10b**, a RAM **10c**, etc. Notably, in the present specification, “ECU” means an electronic control apparatus (electronic control unit) and includes a microcomputer having a CPU, a ROM, a RAM, etc. The CPU realizes various functions by executing instructions stored in the ROM. Furthermore, the driving assistant ECU **10** includes a communication device **10d** which exchanges pieces of information with a server computer via a radio communication line. In addition, the driving assistant ECU **10** includes a timer **10e** for time measurement.

(12) The driving assistant ECU **10** is connected to other ECUs (an engine ECU **31**, a brake ECU **41**, an SBW ECU **51**, and an EPS ECU **61**, which will be described later) via a CAN (controller area network) such that pieces of information can be transmitted and received therebetween.

(13) The on-vehicle sensors **20** include sensors for obtaining pieces of vehicle surrounding information, including information regarding three-dimensional objects present around the vehicle **V** and information regarding division lines on a road surface around the vehicle **V**. Namely, the on-vehicle sensors **20** include, for example, sensors for obtaining pieces of information regarding moving objects such as automobiles (other vehicles), pedestrians, and bicycles and stationary objects such as white lines on the road surface, guard rails, and traffic signals.

(14) Specifically, the on-vehicle sensors **20** include a radar sensor **21**, an ultrasonic sensor **22**, a camera **23**, and a navigation system **24**.

(15) The radar sensor **21** includes a radar transmission/reception section and a signal processing section (not shown). The radar transmission/reception section radiates a radio wave in the millimeter wave band (hereinafter referred to as “millimeter wave”) to a region around the vehicle, and receives a millimeter wave (i.e., reflection wave) reflected by a three-dimensional object present in the region to which the millimeter wave is radiated. The signal processing section computes the distance between the vehicle **V** and the three-dimensional object, the relative speed between the vehicle **V** and the three-dimensional object, the position (direction) of the three-dimensional object in relation to the vehicle **V**, etc. on the basis of the phase difference between the transmitted millimeter wave and the received reflection wave, the level of attenuation of the reflection wave, the time elapsed until the reflection wave is received after the millimeter wave has been transmitted, etc. The signal processing section transmits the computed distance, relative speed, etc. to the driving assistant ECU **10**.

(16) The ultrasonic sensor **22** transmits an ultrasonic wave in the form of pulses to a predetermined region around the vehicle and receives a reflection wave from a three-dimensional object. The ultrasonic sensor **22** computes a “reflection point which is a point on the three-dimensional object at which the transmitted ultrasonic wave is reflected,” the “distance between the ultrasonic sensor

and the three-dimensional object,” etc. on the basis of the time elapsed until the reflection wave is received after the ultrasonic wave has been transmitted. The ultrasonic sensor **22** transmits the computed reflection point, distance, etc. to the driving assistant ECU **10**.

(17) The camera **23** includes an image capturing device and an image analyzing device. The image capturing device is, for example, a digital camera including an image sensor such as a CCD (charge coupled device) or a CIS (CMOS image sensor). The image capturing device is disposed on an upper portion of a front windshield glass. The image capturing device obtains image data by capturing an image of a scene ahead of the vehicle at a predetermined frame rate and outputs the image data to the image analyzing device. The image analyzing device analyzes the obtained image data and obtains, from the image, pieces of information regarding objects located ahead of the vehicle V. The image analyzing device recognizes, for example, white lines (division line, stop line, etc.) on a road and transmits the recognition results to the driving assistant ECU **10**.

(18) The navigation system **24** receives GPS signals from a plurality of artificial satellites and detects the present position (latitude and longitude) of the vehicle V on the basis of the received GPS signals. Also, the navigation system **24** stores map data representing a map. The navigation system **24** transmits vehicle position data to the driving assistant ECU **10**. The vehicle position data represent the detected present position.

(19) The on-vehicle sensors **20** further include sensors for obtaining pieces of information regarding the travel state (speed, acceleration) of the vehicle V and pieces of information regarding the operation states of operating elements provided in the vehicle V.

(20) Specifically, the on-vehicle sensors **20** include a speed sensor **25**, an acceleration sensor **26**, an accelerator pedal sensor **27**, a brake pedal sensor **28**, a shift lever sensor **29**, a steering sensor **2a**, and a switch **2b**.

(21) The speed sensor **25** includes a wheel speed sensor which generates one pulse signal (wheel pulse signal) every time each wheel of the own vehicle rotates a predetermined angle. The speed sensor **25** measures the number of pulses per unit time of the wheel pulse signal transmitted from the wheel speed sensor. The speed sensor **25** computes the rotational speed (wheel speed) of each wheel on the basis of the measured number of pulses, and computes the speed vs (actual speed) of the own vehicle on the basis of the wheel speed of each wheel. The speed sensor **25** transmits data representing the speed vs to the driving assistant ECU **10**.

(22) The acceleration sensor **26** detects acceleration G of the vehicle V (for example, acceleration in the vehicle width direction of the vehicle V, acceleration in the front-rear direction of the vehicle V, etc.). The acceleration sensor **26** transmits data representing the acceleration G to the driving assistant ECU **10**.

(23) The accelerator pedal sensor **27** detects the amount of depression (AD) of an accelerator pedal (not shown) of the vehicle V. The accelerator pedal sensor **27** transmits data representing the amount of depression AD of the accelerator pedal to the driving assistant ECU **10**.

(24) A brake pedal sensor **28** detects the amount of depression (BD) of a brake pedal (not shown) of the vehicle V. The brake pedal sensor **28** transmits data representing the amount of depression BD of the brake pedal to the driving assistant ECU **10**.

(25) The shift lever sensor **29** detects the position of a shift lever (not shown) of the vehicle V (shift lever position SP). The shift lever sensor **29** transmits data representing the shift lever position SP to the driving assistant ECU **10**.

(26) The steering sensor **2a** detects the steering angle θ of the steering wheel. The steering sensor **2a** transmits data representing the detected steering angle θ to the driving assistant ECU **10**.

(27) The switch **2b** is an operating element (for example, a push-button-type switch operating element). A driver operates the switch **2b** for enabling and disabling the risk reduction function. In the case where the driver operates the switch **2b** (pushes a button) in a period during which the risk reduction function is disabled, the switch **2b** transmits to the driving assistant ECU **10** an enablement signal representing that “the driver requests enablement of the risk reduction function.”

When the driving assistant ECU **10** receives the enablement signal, the driving assistant ECU sets the value of a flag F to “1.” In the case where the driver operates the switch **2b** in a period during which the risk reduction function is enabled, the switch **2b** transmits to the driving assistant ECU **10** a disablement signal representing that “the driver requests disablement of the risk reduction function.” When the driving assistant ECU **10** receives the disablement signal, the driving assistant ECU sets the value of the flag F to “0.”

(28) The drive apparatus **30** generates drive power and applies the drive power to drive wheels among all wheels (a front left wheel, a front right wheel, a rear left wheel, and a rear right wheel). The drive apparatus **30** includes an engine ECU **31**, an engine actuator **32**, an internal combustion engine **33**, a transmission **34**, an unillustrated drive power transmission mechanism for transmitting the drive power to the wheels, etc. The engine ECU **31** is connected to the engine actuator **32**. The engine actuator **32** includes a throttle valve actuator for changing the opening degree of a throttle valve of the internal combustion engine **33**. The engine ECU **31** obtains the accelerator pedal depression amount AD from the driving assistant ECU **10**. Notably, the driving assistant ECU **10** can transmit the depression amount AD obtained from the accelerator pedal sensor **27** to the engine ECU **31** after appropriately correcting the depression amount AD. The engine ECU **31** drives the engine actuator **32** in accordance with the depression amount AD obtained from the driving assistant ECU **10**. In this manner, the torque generated by the internal combustion engine **33** is controlled. The torque generated by the internal combustion engine **33** is transmitted to the drive wheels via the transmission **34** and the drive power transmission mechanism (for example, a drive shaft).

(29) Notably, in the case where the vehicle V to which the vehicle control system **1** is applied is a hybrid vehicle (HEV), the engine ECU **31** can control the vehicle drive power generated by either or both of “an internal combustion engine and an electric motor,” which serve as vehicle drive sources. Also, in the case where the vehicle V to which the vehicle control system **1** is applied is a battery electric vehicle (BEV), instead of the engine ECU **31**, a motor ECU may be used so as to control the vehicle drive power generated by “an electric motor,” which serves as a vehicle drive source.

(30) The braking apparatus **40** applies braking forces to the wheels. The braking apparatus **40** includes a brake ECU **41**, a hydraulic circuit **42**, and brake calipers **43**. The hydraulic circuit **42** includes, for example, a reservoir, an oil pump, various valve devices, and an oil pressure sensor, which are not illustrated. Each brake caliper **43** is a hydraulic actuator having a cylinder and a piston. When oil is supplied to the cylinder, the piston is pushed out from the cylinder. A brake pad is provided at the distal end of the piston, and this brake pad is pressed against a brake disc. The brake ECU **41** obtains the brake pedal depression amount BD from the driving assistant ECU **10**. Notably, the driving assistant ECU **10** can transmit the depression amount BD obtained from the brake pedal sensor **28** to the brake ECU **41** after appropriately correcting the depression amount BD. The brake ECU **41** transmits an oil pressure control instruction to the hydraulic circuit **42** in accordance with the depression amount BD obtained from the driving assistant ECU **10**. The hydraulic circuit **42** adjusts the oil pressure within the cylinder of the brake caliper **43** in accordance with the oil pressure control instruction obtained from the brake ECU **41**. In this manner, the braking force applied to the corresponding wheel (brake disc) by the brake caliper **43** is controlled.

(31) The shift change apparatus **50** changes the shift position of the transmission **34**. The shift change apparatus **50** includes an SBW (shift-by-wire) ECU **51**, an SBW actuator **52**, a shift change mechanism **53**, etc. The SBW ECU **51** is connected to the SBW actuator **52**. The SBW ECU **51** obtains the shift lever position SP from the driving assistant ECU **10**. The driving assistant ECU **10** can transmit the shift lever position SP to the SBW ECU **51** after appropriately correcting the shift lever position SP obtained from the shift lever sensor **29**. The SBW ECU **51** transmits a shift change instruction to the SBW actuator **52** in accordance with the shift lever position SP obtained

from the driving assistant ECU **10**. The SBW actuator **52** controls the shift change mechanism **53** in accordance with the shift change instruction obtained from the SBW ECU **51**. In this manner, the shift position of the transmission **34** is changed.

(32) The steering apparatus **60** controls the steering angle of the steerable wheels (the front left wheel and the front right wheel). The steering apparatus **60** includes an electric power steering ECU (hereinafter referred to as the “EPS ECU”) **61**, an assist motor **62**, and a steering mechanism **63**. The EPS ECU **61** is connected to the assist motor **62** (to a drive circuit for the assist motor **62**). The assist motor **62** is built in the steering mechanism **63**. The steering mechanism **63** is a mechanism for steering the steerable wheels. The steering mechanism **63** includes a steering wheel SW, a steering shaft US, an unillustrated steering gear mechanism, etc. The EPS ECU **61** obtains the steering angle θ from the driving assistant ECU **10**. Notably, the driving assistant ECU **10** can transmit the steering angle θ to the EPS ECU **61** after appropriately correcting the steering angle θ obtained from the steering sensor **2a**. The EPS ECU **61** can transmit a steering instruction to the EPS ECU **61** in accordance with the steering angle θ obtained from the driving assistant ECU **10**. Upon receipt of the steering instruction from the driving assistant ECU **10**, the EPS ECU **61** drives the assist motor **62** on the basis of the steering instruction. In this manner, the steering angle of the steerable wheels of the vehicle V is controlled.

(33) (Lane Departure Prevention Function)

(34) Next, the lane departure prevention function (LTA: lane tracing assist) of the vehicle control system **1** will be described. In a state in which the risk reduction function is enabled ($F=1$), the driving assistant ECU **10** computes a target travel line Ld on the basis of pieces of information obtained from the radar sensor **21**, the ultrasonic sensor **22**, and the camera **23**. As shown in FIG. 2, the target travel line Ld is located at a center position between a white line RL (or a curb, a separating zone, or the like) on the right side of a lane in which the vehicle V is travelling and a white line LL (or a curb, a separating zone, or the like) on the left side of the lane. The target travel line Ld extends approximately parallel to the white line RL or the like on the right side of the lane and the white line LL or the like on the left side of the lane. Subsequently, the driving assistant ECU **10** computes a deviation of a center position CL of the front end of the own vehicle from the target travel line Ld (a deviation toward the right side or the left side in the width direction of the lane (hereinafter referred to as an “offset value OFS”)).

(35) Subsequently, the driving assistant ECU **10** issues a predetermined warning to the driver in a state in which the offset value OFS (absolute value) is greater than a threshold value OFS_{th}. Specifically, the driving assistant ECU **10** causes a sound apparatus of the navigation system **24** to reproduce a predetermined warning sound. Furthermore, the driving assistant ECU **10** controls the steering apparatus **60** and optionally the drive apparatus **30** and/or the braking apparatus **40** such that the offset value OFS becomes zero. In the following description, the control (warning/steering/accelerating or decelerating) for preventing departure of the own vehicle from the lane will be referred to as “lane departure prevention control.”

(36) (Emergency Stop Function)

(37) Next, the emergency stop function (EDSS: emergency driving stop system) of the vehicle control system **1** will be described. In the state in which the risk reduction function is enabled ($F=1$), the driving assistant ECU **10** monitors the amount by which each operating element is operated within a predetermined short period of time (a change in the output value of the sensor of each operating element). Namely, the driving assistant ECU **10** repeatedly obtains the accelerator pedal depression amount AD, the brake pedal depression amount BD, the steering wheel steering angle θ , etc. When a state in which none of the operation amounts of all the operating elements changes continues for a threshold time t₁ or more, the driving assistant ECU **10** provides a predetermined warning. Furthermore, when the state in which none of the operation amounts of all the operating elements changes continues for a threshold time t₂ or more from the point of provision of the warning, the driving assistant ECU **10** stops the own vehicle by controlling the

braking apparatus **40**, the steering apparatus **60**, etc. Furthermore, when the state in which none of the operation amounts of all the operating elements changes continues for a threshold time t_3 or more from the point when the own vehicle has been stopped, the driving assistant ECU **10** provides a report (for example, transmits the position information (latitude and longitude) of the own vehicle) to a predetermined server computer via a radio communication line.

(38) (Compulsory Activation Function)

(39) Next, the compulsory activation function of the vehicle control system **1** will be described. When, in a state in which the risk reduction function is disabled ($F=0$), the driving assistant ECU **10** determines on the basis of the pieces of information obtained from the on-vehicle sensors **20** that, as shown in FIG. **3**, the own vehicle has departed from the lane (on which the vehicle has traveled for a predetermined period of time or longer) and lightly collided with an object (for example, a guard rail or the like) (occurrence of a collision which is light so that an airbag apparatus does not operate), the driving assistant ECU **10** forcibly enables the risk reduction function ($F=1$). Namely, the driving assistant ECU **10** switches the value of the flag F from “0” to “1” despite of the switch **2b** being not operated (compulsory activation). The driving assistant ECU **10** determines that the own vehicle has lightly collided with an object, for example, when a condition X described below is satisfied.

(40) (Condition X).

(41) The magnitude of the acceleration G falls within a predetermined range G_A , and the integral value of the magnitude of the acceleration G has exceeded a threshold value, the integral value of the magnitude of the acceleration being that over a predetermined time from a point in time when the magnitude of the acceleration G has exceeded the lower limit (>0) of the range G_A .

(42) Notably, the range G_A is determined beforehand as a range of acceleration which is greater than accelerations occurring in ordinary driving states (sudden deceleration (forward G), sudden acceleration (rearward G), sudden turn (lateral G), and vibrations due to road surface (vertical G)) and which can be considered to occur as a result of light collision.

(43) Furthermore, after forcibly enabling the risk reduction function, the driving assistant ECU **10** obtains the speed v_s from the speed sensor **25**. In the case where the speed v_s is higher than a threshold value v_{sth} (namely, in the case where the own vehicle has hit against an object at a relatively high speed), the driving assistant ECU **10** controls the braking apparatus **40**, the steering apparatus **60**, etc. such that the own vehicle stops immediately. In this case, regardless of whether or not the driver is operating an operating element(s), the driving assistant ECU **10** forcibly stops the own vehicle (compulsory stop function). The priority of this compulsory stoppage function is higher than those of the lane departure prevention function and the emergency stop function.

(44) Notably, in the case where the acceleration G exceeds the upper limit of the range G_A (the impact acting on the own vehicle is very large), the driving assistant ECU **10** (or another ECU) controls the braking apparatus **40**, the steering apparatus **60**, etc. such that the own vehicle stops immediately.

(45) Next, operation of the CPU **10a** of the driving assistant ECU **10** (hereinafter referred to simply as the “CPU”) (programs **P1**, **P2**, and **P3** which realize the lane departure prevention function, the emergency stop function, and the compulsory activation function, respectively) will be described with reference to FIGS. **4** to **6**. During a period during which the engine of the own vehicle is operating (an ignition switch is in an ON state), the CPU executes the programs **P1**, **P2**, and **P3** at predetermined intervals.

(46) (Program **P1**)

(47) The CPU starts execution of the program **P1** from step **100** and proceeds to step **101**.

(48) When the CPU proceeds to step **101**, the CPU determines whether or not the risk reduction function is enabled (whether or not the value of the flag F is “1”). In the case where the risk reduction function is enabled (**101**: Yes), the CPU proceeds to step **102**. Meanwhile, in the case where the risk reduction function is disabled (**101**: No), the CPU proceeds to step **105** and ends the

execution of the program P1.

(49) When the CPU proceeds to step **102**, the CPU computes the offset value OFS. Subsequently, the CPU proceeds to step **103**.

(50) When the CPU proceeds to step **103**, the CPU determines whether or not the offset value OFS (absolute value) is equal to or smaller than a threshold value OFSth. In the case where the offset value OFS is equal to or smaller than the threshold value OFSth (**103**: Yes), the CPU proceeds to step **105**. Meanwhile, in the case where the offset value OFS is larger than the threshold value OFSth (**103**: No), the CPU proceeds to step **104**.

(51) When the CPU proceeds to step **104**, the CPU executes lane departure prevention control (warning/steering/accelerating or decelerating). The CPU then proceeds to step **105**.

(52) (Program P2)

(53) The CPU starts execution of the program P2 from step **200** and proceeds to step **201**. When the CPU proceeds to step **201**, the CPU determines whether or not the risk reduction function is enabled (whether or not the value of the flag F is “1”). In the case where the risk reduction function is enabled (**201**: Yes), the CPU proceeds to step **202**. Meanwhile, in the case where the risk reduction function is disabled (**201**: No), the CPU proceeds to step **215** and ends the execution of the program P2.

(54) When the CPU proceeds to step **202**, the CPU initializes a measured time ta (output value) of the timer **10e** to “0” and causes the timer **10e** to start time measurement. Subsequently, the CPU proceeds to step **203**.

(55) When the CPU proceeds to step **203**, the CPU determines whether or not the driver is operating any of the operating elements (whether or not the operation amounts of all the operating elements (changes in the output values of the sensors) are “0”). In the case where the driver is operating none of the operating elements (the driver is suffering from an abnormal condition (**203**: Yes)), the CPU proceeds to step **204**. Meanwhile, in the case where the driver is operating any of the operating elements (**203**: No), the CPU proceeds to step **215**.

(56) When the CPU proceeds to step **204**, the CPU determines whether or not the measured time ta is greater than a threshold value t1. In the case where the measured time ta is greater than the threshold value t1 (**204**: Yes), the CPU proceeds to step **205**. Meanwhile, in the case where the measured time ta is equal to or less than the threshold value t1, the CPU returns to step **203**.

(57) When the CPU proceeds to step **205**, the CPU causes the navigation system **24** to reproduce a predetermined warning sound. Subsequently, the CPU proceeds step **206**.

(58) When the CPU proceeds to step **206**, the CPU initializes the measured time ta (output value) of the timer **10e** to “0” and again causes the timer **10e** to start time measurement. Subsequently, the CPU proceeds to step **207**.

(59) When the CPU proceeds to step **207**, the CPU determines whether or not the driver is operating any of the operating elements (whether or not the operation amounts of all the operating elements are “0”). In the case where the driver is operating none of the operating elements (**207**: Yes), the CPU proceeds to step **208**. Meanwhile, in the case where the driver is operating any of the operating elements (**207**: No), the CPU proceeds to step **215**.

(60) When the CPU proceeds to step **208**, the CPU determines whether or not the measured time ta is greater than a threshold value t2. In the case where the measured time ta is greater than the threshold value t2 (**208**: Yes), the CPU proceeds to step **209**. Meanwhile, in the case where the measured time ta is equal to or less than the threshold value t2, the CPU returns to step **207**.

(61) When the CPU proceeds to step **209**, the CPU brakes and stops the own vehicle (emergency stop). Namely, while flashing the hazard lights, the CPU controls the braking apparatus **40** and the steering apparatus **60** such that the own vehicle moves to a safe region (for example, a shoulder or a lane where the traffic volume is small), while decelerating, and stops there. Subsequently, the CPU proceeds to step **210**. Notably, the CPU executes step **209** preferentially over the program P1.

(62) When the CPU proceeds to step **210**, the CPU initializes the measured time ta (output value)

of the timer **10e** to “0” and again causes the timer **10e** to start time measurement. Subsequently, the CPU proceeds to step **211**.

(63) When the CPU proceeds to step **211**, the CPU determines whether or not the driver is operating any of the operating elements (whether or not the operation amounts of all the operating elements are “0”). In the case where the driver is operating none of the operating elements (**211: Yes**), the CPU proceeds to step **212**. Meanwhile, in the case where the driver is operating any of the operating elements (**211: No**), the CPU proceeds to step **215**.

(64) When the CPU proceeds to step **212**, the CPU determines whether or not the measured time t_a is greater than a threshold value t_3 . In the case where the measured time t_a is greater than the threshold value t_3 (**212: Yes**), the CPU proceeds to step **213**. Meanwhile, in the case where the measured time t_a is equal to or less than the threshold value t_3 , the CPU returns to step **211**.

(65) When the CPU proceeds to step **213**, the CPU moves the shift position to a parking position and transmits (reports) a predetermined piece of information to the server computer. Subsequently, the CPU proceeds to step **214**.

(66) When the CPU proceeds to step **214**, the CPU sets the value of the flag F to “0.” As a result, repeated issuance of the warning and the report is prevented. Subsequently, the CPU proceeds to step **215**.

(67) (Program **P3**)

(68) The CPU starts execution of the program **P3** from step **300** and proceeds to step **301**.

(69) When the CPU proceeds to step **301**, the CPU determines whether or not the risk reduction function is disabled (whether or not the value of the flag F is “0”). In the case where the risk reduction function is disabled (**301: Yes**), the CPU proceeds to step **302**. Meanwhile, in the case where the risk reduction function is enabled (**301: No**), the CPU proceeds to step **312** and ends the execution of the program **P3**.

(70) When the CPU proceeds step **302**, the CPU determines, on the basis of the information obtained from the camera **23**, whether or not the own vehicle has departed from a lane (whether or not the own vehicle is crossing the dividing line between that lane and a lane adjacent thereto). In the case where the own vehicle has departed from that lane (**302: Yes**), the CPU proceeds to step **303**. Meanwhile, in the case where the own vehicle has not departed from that line, the CPU proceeds to step **312**.

(71) When the CPU proceeds to step **303**, the CPU determines whether or not the own vehicle has lightly collided with an object (whether or not the condition X is satisfied). In the case where the own vehicle has lightly collided with an object (**303: Yes**), the CPU proceeds to step **304**. Meanwhile, in the case where the own vehicle has not lightly collided with an object (**303: No**), the CPU proceeds to step **312**. Notably, in the case where the impact acting on the own vehicle is very large (the case where the acceleration G exceeds the upper limit of the range G_A), the driving assistant ECU **10** immediately stops the own vehicle by executing a program (not shown) whose priority is higher than that those of the program **P1** to the program **P3**.

(72) When the CPU proceeds to step **304**, the CPU sets the value of the flag F to “1” (compulsory activation). Subsequently, the CPU proceeds to step **305**.

(73) When the CPU proceeds to step **305**, the CPU determines whether or not the speed v_s of the own vehicle is greater than a threshold value v_{sth} . In the case where the speed v_s is greater than the threshold value v_{sth} (**305: Yes**), the CPU proceeds to step **306**. Meanwhile, in the case where the speed v_s is equal to or less than the threshold value v_{sth} (**305: No**), the CPU proceeds to step **312**. In this case, by the lane departure prevention function forcibly enabled, the drive apparatus **30**, the braking apparatus **40**, the steering apparatus **60**, etc. are controlled such that the own vehicle travels along the lane. When the state in which the driver does not operate any of the operating elements continues after that, the own vehicle is stopped by the emergency stop function. Notably, since step **305** is executed immediately after the risk reduction function has been forcibly enabled, at this point in time, braking by the emergency stop function (step **209**) has not yet been started.

(74) When the CPU proceeds to step **306**, the CPU brakes and stops the own vehicle (compulsory stop). Namely, while flashing the hazard lights, the CPU controls the braking apparatus **40**, the steering apparatus **60**, etc. such that the own vehicle moves to a safe region (for example, a shoulder or a lane where the traffic volume is small), while decelerating, and stops there. Subsequently, the CPU proceeds to step **307**. Notably, the CPU executes step **306** preferentially over the program **P1** and the program **P2**.

(75) When the CPU proceeds to step **307**, the CPU initializes the measured time *tb* (output value) of the timer **10e** to “0” and causes the timer **10e** to start time measurement. Subsequently, the CPU proceeds to step **308**.

(76) When the CPU proceeds to step **308**, the CPU determines whether or not the driver is operating any of the operating elements (whether or not the operation amounts of all the operating elements are “0”). In the case where the driver is operating none of the operating elements (**308**: Yes), the CPU proceeds to step **309**. Meanwhile, in the case where the driver is operating any of the operating elements (**308**: No), the CPU proceeds to step **312**.

(77) When the CPU proceeds to step **309**, the CPU determines whether or not the measured time *tb* is greater than a threshold value *t3*. In the case where the measured time *tb* is greater than the threshold value *t3* (**309**: Yes), the CPU proceeds to step **310**. Meanwhile, in the case where the measured time *tb* is equal to or less than the threshold value *t3*, the CPU returns to step **308**.

(78) When the CPU proceeds to step **310**, the CPU moves the shift position to the parking position and transmits (reports) a predetermined piece of information to the server computer. Subsequently, the CPU proceeds to step **312**.

(79) When the CPU proceeds to step **311**, the CPU sets the value of the flag *F* to “0.” Subsequently, the CPU proceeds to step **312**.

(80) (Effects)

(81) In a vehicle on which the above-described vehicle control system **1** is mounted, a driver can manually enable and disable the risk reduction function. When the own vehicle departs from a lane and lightly collides with an object in a state in which the risk reduction function has been manually disabled, the risk reduction function is forcedly enabled. As a result, for example, in a situation where it has become difficult for the driver to drive and operate the own vehicle, the driving and operation are assisted by the driving assistant ECU **10** (LTA). Also, in a situation where the driver has become unable to drive and operate the own vehicle, the own vehicle is stopped by the driving assistant ECU **10** (EDSS). Accordingly, the vehicle control system **1** can enhance safety of the vehicle as compared with the conventional system.

(82) Also, in the case where the own vehicle has lightly collided with an object at a relatively high speed, the own vehicle is stopped immediately (step **305** to step **311** of FIG. **6**). As a result, safety of the vehicle can be enhanced further.

(83) The present invention is not limited to the above-described embodiment, and, as will be described below, various modifications can be employed without departing from the scope of the present invention.

(84) <Modification 1>

(85) In the above-described embodiment, when the driver pushes the switch **2b**, switching is performed between the state in which both the lane departure prevention function and the emergency stop function are on (are enabled) and the state in which both the lane departure prevention function and the emergency stop function are off (are disabled). Alternatively, the vehicle control system may be configured such that the driver can individually turn on and off the lane departure prevention function and the emergency stop function. For example, there are provided a switch **2b1** for turning the lane departure prevention function on and off and a switch **2b2** for turning the emergency stop function on and off. In this case, instead of the flag *F*, flags *F1* and *F2* are used. When the lane departure prevention function is on, the value of the flag *F1* is “1,” and when the lane departure prevention function is off, the value of the flag *F1* is “0.” Similarly,

when the emergency stop function is on, the value of the flag F2 is “1,” and when the emergency stop function is off, the value of the flag F2 is “0.”

(86) In the case, when the CPU determines in step **101** of FIG. **4** that the value of the flag F1 is “1,” the CPU proceeds to step **102**, and, when the CPU determines in step **101** that the value of the flag F1 is “0,” the CPU proceeds to step **105**. Also, when the CPU determines in step **201** of FIG. **5** that the value of the flag F2 is “1,” the CPU proceeds to step **202**, and, when the CPU determines in step **201** that the value of the flag F2 is “0,” the CPU proceeds to step **215**. Also, the CPU sets the value of the flag F2 to “0” in step **214** of FIG. **5**. Also, in this case, when the CPU determines in step **301** of FIG. **6** that at least one of the values of the flags F1 and F2 is “0,” the CPU proceeds to step **302**, and, when the CPU determines in step **301** that both the values of the flags F1 and F2 are “1,” the CPU proceeds to step **312**. Also, in step **304**, the CPU sets the values of the flags F1 and F2 to “1.” Furthermore, in step **311**, the CPU sets the values of the flags F1 and F2 to “0.”

(87) <Modification 2>

(88) In the above-described embodiment, the CPU determines whether or not the driver is suffering from an abnormal condition by determining whether or not the driver is operating any of the operating elements, and determines, on the basis of the result of the determination, whether to perform braking of the own vehicle and provide a report. Instead of or in addition to the above determination, the CPU may perform the following determination. A camera for photographing the driver is provided in the cabin, and the CPU determines, on the basis of an image obtained from the camera, whether or not the driver is suffering from an abnormal condition. In this case, the CPU is programed to allow the driver to manually perform switching between a state in which a function of determining the state of the driver on the basis of the image of the driver (hereinafter referred to as the “driver image determination function”) is enabled and a state in which the driver image determination function is disabled. In the case where the own vehicle departs from a lane and lightly collides with an object in the state in which the driver image determination function has been disabled, the CPU forcibly enables the driver image determination function.

(89) Notably, the vehicle control system **1** may include an apparatus which automatically disables the lane departure prevention function upon detection that the driver has intentionally performed a lane change, an apparatus which automatically enables the lane departure prevention function upon detection that the driver is suffering from an abnormal condition, etc.

(90) The vehicle V may be an autonomous vehicle.

Claims

1. A vehicle control system comprising: an on-vehicle sensor for outputting a piece of information regarding an object around an own vehicle and a piece of information regarding a travel state of the own vehicle; a control apparatus which has, as a risk reduction function, at least one of a lane departure prevention function and an emergency stop function, the lane departure prevention function controlling at least one of a warning apparatus, a drive apparatus, a braking apparatus, and a steering apparatus of the own vehicle, on the basis of the pieces of information obtained from the on-vehicle sensor, such that departure of the own vehicle from a lane in which the own vehicle is travelling is prevented, the emergency stop function stopping the own vehicle when a driver is suffering from an abnormal condition; and a switching apparatus for enabling and disabling the risk reduction function, wherein the control apparatus is configured to determine whether the own vehicle has lightly collided with an object, wherein the control apparatus determines that the own vehicle has lightly collided with the object in the case that an integral value of a magnitude of acceleration of the own vehicle has exceeded a threshold value, the integral value of the magnitude of acceleration being that during a predetermined time from a point in time when (i) the magnitude of the acceleration has exceeded a lower limit of 0 of a predetermined range in which an airbag apparatus does not operate and (ii) has not exceeded an upper limit of the predetermined range in

which the airbag apparatus does operate, and based upon the determination that the own vehicle has lightly collided with the object, forcedly enable the risk reduction function when the own vehicle collides with an object in a state in which the risk reduction function has been disabled by the switching apparatus.

2. The vehicle control system according to claim 1, wherein the switching apparatus includes a first operating element for manually enabling and disabling the lane departure prevention function and a second operating element for manually enabling and disabling the emergency stop function, and in the case where the own vehicle departs from the lane and collides with an object in a state in which at least one of the lane departure prevention function and the emergency stop function has been manually disabled, the control apparatus forcedly enables the function disabled manually.

3. The vehicle control system according to claim 1, wherein the risk reduction function is the emergency stop function, and the control apparatus controls the braking apparatus so as to immediately stop the own vehicle, irrespective of whether or not the driver is suffering from an abnormal condition, in the case where the speed of the own vehicle exceeds a threshold value in a period between a point in time when the control apparatus has forcedly enabled the risk reduction function and a point in time when braking of the own vehicle by the emergency stop function is started.

4. The vehicle control system according to claim 2, wherein the risk reduction function is the emergency stop function, and the control apparatus controls the braking apparatus so as to immediately stop the own vehicle, irrespective of whether or not the driver is suffering from an abnormal condition, in the case where the speed of the own vehicle exceeds a threshold value in a period between a point in time when the control apparatus has forcedly enabled the risk reduction function and a point in time when braking of the own vehicle by the emergency stop function is started.

5. The vehicle control system according to claim 1, wherein a state in which the own vehicle lightly collides with the object is a state in which the magnitude of the acceleration is greater than the magnitude of the acceleration that occurs within an ordinary driving state, and less than the magnitude of the acceleration that occurs when an impact on the own vehicle causes the airbag to operate.
